THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 12: **Surface Areas and Volumes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A tent is a cylinder of radius 3.5 m and height 4 m surmounted by a cone of the same radius and height 12 m. The area of canvas (in m^2) needed to make the tent is: 1

(a) 88 m^2 **(b)** 137.5 m^2 **(c)** 220 m^2 **(d)** 225.5 m^2

2. A hemispherical bowl has outer radius 6 cm and inner radius 3 cm. The volume of material used to make the bowl is: 1

(a) 396 cm^3 **(b)** 792 cm^3 **(c)** 1188 cm^3 **(d)** 198 cm^3

3. A hemispherical depression of diameter 21 cm, the largest possible, is scooped out of the top face of a cubical block of edge 21 cm. The volume of the remaining block is: 1

(a) 2425.5 cm^3 **(b)** 4410 cm^3 **(c)** 6835.5 cm^3 **(d)** 9261 cm^3

4. A solid metallic sphere of radius 12 cm is melted and drawn into a wire of circular cross-section of radius 3 cm. The length of the wire is: 1

(a) 768 cm **(b)** 128 cm **(c)** 256 cm **(d)** 512 cm

5. A solid sphere of radius 3 cm is completely dropped into water in a cylindrical vessel of base radius 6 cm, and no water spills. The rise in the water level is: 1

(a) 2 cm **(b)** 0.5 cm **(c)** 1 cm **(d)** 3 cm

6. A capsule is a cylinder of radius 7 cm and length 10 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 1056 cm^2 **(b)** 1364 cm^2 **(c)** 748 cm^2 **(d)** 2112 cm^2

7. A toy is a cone of base radius 42 cm and height 56 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 25872 cm^2 **(b)** 20328 cm^2 **(c)** 9240 cm^2 **(d)** 18480 cm^2

8. A solid metal cuboid 24 cm by 12 cm by 18 cm is melted and recast into cubes of edge 3 cm. The number of cubes obtained is: 1

(a) 191 **(b)** 192 **(c)** 576 **(d)** 384

9. A tent is a cylinder of radius 3.5 m and height 10 m surmounted by a cone of the same radius and height 12 m. The area of canvas (in m^2) needed to make the tent is: 1

(a) 357.5 m^2 **(b)** 220 m^2 **(c)** 137.5 m^2 **(d)** 352 m^2

10. A hemispherical bowl has outer radius 8 cm and inner radius 2 cm. The volume of material used to make the bowl is: 1

(a) 2112 cm^3 **(b)** 3168 cm^3 **(c)** 528 cm^3 **(d)** 1056 cm^3

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A solid metallic cone of base radius 6 cm and height 3 cm is melted and recast into a solid sphere. Find the radius of the sphere. 2

12. A solid metallic cylinder of base radius 6 cm and height 48 cm is melted and recast into solid spheres of radius 3 cm. Find the number of spheres obtained. 2

13. A solid is a cube of edge 35 cm surmounted by a hemisphere of diameter 35 cm (the hemisphere rests on a face of the cube). Find the total surface area of the solid. 2

14. A solid metallic cone of base radius 3 cm and height 96 cm is melted and recast into a solid sphere. Find the radius of the sphere. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A capsule is a cylinder of radius 10.5 cm and length 12 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

16. A toy is a cone of base radius 21 cm and height 14 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

17. A wooden article is made by scooping out a hemisphere of radius 3.5 cm from each end of a solid cylinder of height 15 cm and base radius 3.5 cm. Find the total surface area of the article. 3

OR

A capsule is a cylinder of radius 10.5 cm and length 18 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

18. A toy is a cone of base radius 21 cm and height 42 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

OR

A wooden article is made by scooping out a hemisphere of radius 7 cm from each end of a solid cylinder of height 10 cm and base radius 7 cm. Find the total surface area of the article. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A conical vessel of radius 6 cm and height 20 cm is full of water. The water is poured into a cylindrical vessel of base radius 4 cm. Find the height of the water in the cylindrical vessel. 5

OR

Water from a full cylindrical tank of radius 7 cm and height 20 cm is poured into an empty rectangular tank of base 22 cm by 14 cm. Find the height of water in the rectangular tank. 5

20. A conical vessel of radius 6 cm and height 16 cm is full of water. The water is poured into a cylindrical vessel of base radius 4 cm. Find the height of the water in the cylindrical vessel. 5

OR

Water from a full cylindrical tank of radius 7 cm and height 30 cm is poured into an empty rectangular tank of base 21 cm by 11 cm. Find the height of water in the rectangular tank. 5

END OF QUESTION PAPER — Total: 40 Marks